Hackathon for Human-Centered Design of Large Language Model Interfaces
ACM SIGCHI Supported Hackathon

Design Document

Project Title: Negotiating AI Memory

Problem Statement ID: PS06

Team Name: HalluciNot

Team Members

Veerbhadra Mahant
First author's affiliation, Pimpri Chinchwad College of Engineering, veerbhadra.mahant24@pccoepune.org
Ved Dhapodkar
Second author's affiliation, Pimpri Chinchwad College of Engineering, ved.dhapodkar24@pccoepune.org
Tanmayee Gaonkar
Third author's affiliation, Pimpri Chinchwad College of Engineering, tanmayee.gaonkar24@pccoepune.org
Shruti Jahagirdar
Third author's affiliation, Pimpri Chinchwad College of Engineering, shruti.jahagirdar24@pccoepune.org

Abstract (200-250 words)

Keywords: 3-5 keywords

1. Introduction (background & motivation)

1.1 Problem Statement

Select one of the official problem statements.

1.2 Problem Description

Briefly explain:

- What problem are you solving?
- Why is it important?
- Who is affected?

2. Literature Review

2.1 Persistent AI Memory, Personalization Paradoxes, and User Mental Models in Conversational LLMs

(synthesized from Chen et al., 2026, "Relational Gains, Privacy Strains," CHI '26)

Divergent Mental Models: Human Memory vs. Algorithmic Persistence. Human conversational memory is selective and lossy — details fade, are reframed, or are dropped entirely over time. Chen et al.'s participants, by contrast, encountered a system whose memory behaves in the opposite direction: exhaustive, precise, and durable by default. Several participants reported being unaware their memory feature was active at all, and comparably unaware that stored entries could be viewed or deleted — a gap that only became visible when the system surfaced retained details with a granularity (down to specific times and locations) that a human conversational partner would never retain. This asymmetry suggests users approach AI memory with a folk model borrowed from human forgetting, and

are repeatedly recalibrated — often unpleasantly — by an underlying architecture built for unforgetting accuracy rather than graceful decay.

Expectancy Violations and the Personalization–Privacy Paradox. This mismatch between assumed and actual persistence maps onto Expectancy Violations Theory: when a system either fails to recall something a user expected it to retain, or recalls something the user expected it to have discarded, the resulting surprise is read negatively, as an intrusion rather than a convenience. Chen et al.'s participants also described genuine uncertainty about *how* retained information was processed, retrieved, and shared — an opacity that becomes more pronounced across model versions or sessions, where users cannot verify whether "forgetting" a fact in one context actually removes it elsewhere. This uncertainty activates a privacy calculus: users must weigh the relational and functional benefits of personalization against a cost they cannot fully see or audit. A related concern is unwanted profiling — participants described discomfort not with memory itself, but with the aggregate self-image the system had assembled from their inputs, an image they had no say in composing and were not sure they endorsed.

Relational Affordances vs. Agency: Designing Transparent and Bounded Memory. Chen et al. translate these findings into concrete design suggestions: live, always-visible indicators of what is being remembered; time-limited retention rather than indefinite storage; support for multiple context-specific personas (so a "work" self and a "personal" self aren't flattened into one profile); and periodic, user-facing prompts to confirm, correct, or delete the assumptions memory has accumulated. Underlying these suggestions is a tension the authors term "bounded friendship" — memory makes an AI feel more relationally present, which is itself powered by machine heuristics (the tendency to extend human-like social trust to a system that appears to know and remember you), but that same relational pull is precisely what makes unchecked memory risky. Good design, in this framing, is not about maximizing memory or minimizing it, but about keeping the system's recall legible and revisable enough that the relationship stays negotiated rather than assumed.

2.2 Guidelines for Human-AI Interaction

		AI Design Guidelines	Example Applications of Guidelines
Initially	G1	**Make clear what the system can do.** Help the user understand what the AI system is capable of doing.	[Activity Trackers, Product #1] "Displays all the metrics that it tracks and explains how. Metrics include movement metrics such as steps, distance traveled, length of time exercised, and all-day calorie burn, for a day."
	G2	**Make clear how well the system can do what it can do.** Help the user understand how often the AI system may make mistakes.	[Music Recommenders, Product #1] "A little bit of hedging language: 'we think you'll like'."
During interaction	G3	**Time services based on context.** Time when to act or interrupt based on the user's current task and environment.	[Navigation, Product #1] "In my experience using the app, it seems to provide timely route guidance. Because the map updates regularly with your actual location, the guidance is timely."
	G4	**Show contextually relevant information.** Display information relevant to the user's current task and environment.	[Web Search, Product #2] "Searching a movie title returns show times in near my location for today's date"
	G5	**Match relevant social norms.** Ensure the experience is delivered in a way that users would expect, given their social and cultural context.	[Voice Assistants, Product #1] "[The assistant] uses a semi-formal voice to talk to you - spells out "okay" and asks further questions."
	G6	**Mitigate social biases.** Ensure the AI system's language and behaviors do not reinforce undesirable and unfair stereotypes and biases.	[Autocomplete, Product #2] "The autocomplete feature clearly suggests both genders [him, her] without any bias while suggesting the text to complete."
When wrong	G7	**Support efficient invocation.** Make it easy to invoke or request the AI system's services when needed.	[Voice Assistants, Product #1] "I can say [wake command] to initiate."
	G8	**Support efficient dismissal.** Make it easy to dismiss or ignore undesired AI system services.	[E-commerce, Product #2] "Feature is unobtrusive, below the fold, and easy to scroll past...Easy to ignore."
	G9	**Support efficient correction.** Make it easy to edit, refine, or recover when the AI system is wrong.	[Voice Assistants, Product #2] "Once my request for a reminder was processed I saw the ability to edit my reminder in the UI that was displayed. Small text underneath stated 'Tap to Edit' with a chevron indicating something would happen if I selected this text."
	G10	**Scope services when in doubt.** Engage in disambiguation or gracefully degrade the AI system's services when uncertain about a user's goals.	[Autocomplete, Product #1] "It usually provides 3-4 suggestions instead of directly auto completing it for you"
	G11	**Make clear why the system did what it did.** Enable the user to access an explanation of why the AI system behaved as it did.	[Navigation, Product #2] "The route chosen by the app was made based on the Fastest Route, which is shown in the subtext."
Over time	G12	**Remember recent interactions.** Maintain short term memory and allow the user to make efficient references to that memory.	[Web Search, Product #1] "[The search engine] remembers the context of certain queries, with certain phrasing, so that it can continue the thread of the search (e.g., 'who is he married to' after a search that surfaces Benjamin Bratt)"
	G13	**Learn from user behavior.** Personalize the user's experience by learning from their actions over time.	[Music Recommenders, Product #2] "I think this is applied because every action to add a song to the list triggers new recommendations."
	G14	**Update and adapt cautiously.** Limit disruptive changes when updating and adapting the AI system's behaviors.	[Music Recommenders, Product #2] "Once we select a song they update the immediate song list below but keeps the above one constant."
	G15	**Encourage granular feedback.** Enable the user to provide feedback indicating their preferences during regular interaction with the AI system.	[Email, Product #1] "The user can directly mark something as important, when the AI hadn't marked it as that previously."
	G16	**Convey the consequences of user actions.** Immediately update or convey how user actions will impact future behaviors of the AI system.	[Social Networks, Product #2] "[The product] communicates that hiding an Ad will adjust the relevance of future ads."
	G17	**Provide global controls.** Allow the user to globally customize what the AI system monitors and how it behaves.	[Photo Organizers, Product #1] "[The product] allows users to turn on your location history so the AI can group photos by where you have been."
	G18	**Notify users about changes.** Inform the user when the AI system adds or updates its capabilities.	[Navigation, Product #2] "[The product] does provide small in-app teaching callouts for important new features. New features that require my explicit attention are pop-ups."

This work organizes AI-UX guidance into four phases: initial setup, during interaction, when the system errs, and over time. Several guidelines are directly load-bearing for a memory-negotiation interface: supporting efficient correction when the system is wrong (G9), providing global controls over what is monitored and how (G17), conveying to users how their actions will affect future system behavior (G16), and updating or adapting cautiously rather than making abrupt, disruptive changes to personalization (G14). Read alongside Chen et al., these guidelines function as a general-purpose checklist against which a memory interface's transparency and controllability can be evaluated, independent of any one product.

2.3 The AI Memory Gap: Users Misremember What They Created With AI or Without

This paper examines a different but related failure mode: not what the *system* remembers, but what the *user* misremembers about their own co-creative process. The central finding is that using an AI collaborator measurably weakens people's ability to later recall the source and outcome of their own creative work. The authors trace this to source-memory failure: when memory of who-contributed-what is weak, people fall back on heuristic cues — abstract, half-formed ideas tend to feel self-generated (especially if the user went on to elaborate them further), while fluent, fully-formed sentence-level text tends to feel externally sourced, biasing attribution toward the AI even when the reverse was true. Their design recommendation is to embed distinguishing perceptual cues during generation — different colors, animations, or interaction modes for self-authored versus AI-authored content — and to preserve interaction traces (e.g., prompt history) so users can later reconstruct or verify their own contribution. The authors frame this as a matter of responsible co-creation design: memory failures in human-AI collaboration are not a side effect but a design problem to actively mitigate.

2.4 RECALLbot: Designing Agentic Memory and Reciprocal Disclosure for Human-Chatbot Relationships

RECALLbot frames memory as the mechanism through which chatbots achieve personalization, behavioral consistency, and — most importantly — a sense of identity continuity that underpins trust and relationship formation with users. The authors decompose memory mechanisms into four stages: extraction (deriving preferences and facts from interaction logs), representation, reference (using stored memory in later generation), and updating. Their findings reinforce a recurring theme across this review: users want to see their memory directly and retain active control over it, and the same mechanisms that enable personalization simultaneously raise trust, security, and privacy concerns due to limited explainability and the risk of exposing personal information. The authors conclude that agentic memory construction and updating are not optional extras but essential components of a functioning social chatbot.

2.5 Design Principles for Generative-AI Applications

This paper proposes three design goals (DG1–DG3) that closely prefigure the aims of the current project. DG1 calls for constructing "agentic memories" with a human-like internal structure — background memories that establish identity, everyday memories that sustain it, and relational memories that connect to the user — expressed in affectively human ways (referencing feeling, referencing the external world) to reduce the perceived distance between user and system. DG2 calls for memory disclosure strategies that stay topically relevant and adapt to conversational pacing, alternating between the system disclosing its own "memories" and inviting the user to disclose theirs. DG3 calls for intuitive, low-friction controls over what is remembered, how disclosure is paced, and which topics are in scope. Of the five papers reviewed, this one is the closest structural analogue to "Negotiating AI Memory" — DG2 and DG3 in particular already frame memory as a two-way, in-conversation negotiation rather than a static settings panel.

Synthesis

Across all five papers, a consistent gap emerges: memory in conversational AI is treated either as a backend extraction-and-storage problem (RECALLbot, Design Principles) or as a source of user-side confusion and misattribution (Chen et al., AI Memory Gap), with design guidance (Guidelines for Human-AI Interaction) sitting above both as general best practice. None of them fully operationalize memory as a *negotiated, ongoing exchange* between user and system — visible, correctable, and mutually paced in real time — which is the gap this project positions itself to address.

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3. Proposed Solution

Main idea: Instead of letting an AI assistant accumulate memory silently and exposing it only through an after-the-fact settings page, our system treats memory formation as a **real-time negotiation** between the user and the system. Every fact the model would remember is surfaced *at the moment it forms* — not audited later — and the user accepts, edits, rescopes, or discards it inline before it ever persists. Memory items also carry an **assertion status** (in_progress, completed, planned, abandoned, hypothetical, third_party) rather than being stored as flat, timeless facts — so "working on a paper" cannot silently calcify into "wrote a paper" months later.

Key features:

- **Negotiation loop** (surface → decide → scope) — candidate memories appear as inline review cards the user can accept, edit, or reject before storage
- **Stated vs. inferred distinction**, never conveyed by color alone (fill, border style, glyph, and a mono label all carry it — accessibility-safe)
- **Session vs. persistent scoping** — "use this now, don't keep it" is a structural guarantee: the extraction pass never runs on an off-the-record turn, not just hidden in the UI
- **Use-time attribution** — every AI response shows which memories shaped it, and any one can be revoked with the answer regenerated to show the difference
- **Stake-proportional verification** — before a high-stakes artifact (e.g., a CV) is produced, every memory-derived claim is checked in code against its actual assertion status, surfacing unconfirmed claims instead of asserting them
- **Provenance-aware deletion** — deleting a memory shows what else it affects, as text, before you confirm
- **Pre-send PII intervention** — irreversible data (card numbers, credentials) is caught client-side before it ever leaves the device
- **Consent that actually gates behavior** — declining memory immediately flips every message to off-the-record, verified end-to-end on the network layer, not just in the UI

Why our solution is different: Most memory UIs are a dashboard bolted onto an already-silent extraction pipeline — inspection after the fact, with no say at the moment memory forms. We reject that shape on purpose (see our anti-goals: no dashboard-as-headline-artifact, no graph view that just looks impressive). Three things set this apart: (1) negotiation happens *before* persistence, not after; (2) memory has a **temporal-validity model** (assertion status) instead of being treated as permanently true the moment it's written — this is what catches the "stale paper on a CV" failure mode that plain accuracy checks miss; (3) claims like "you can keep it session-only" are proven structurally (a live database query returning zero rows), not merely asserted in a settings toggle.

4. Target Users

- Primary users : **the heavy personal user.** Someone who uses one AI assistant across both work and personal life. They benefit most from persistent memory and are exposed most by it — they want continuity without surveillance.
- Secondary users : **the cautious professional.** Someone who uses AI mainly for work output and cares less about privacy in the abstract than about *accuracy in artifacts* the AI produces on their behalf. A stale or wrong memory surfacing in a CV or report is their nightmare scenario, not a minor annoyance

Example:

Attribute	Description
Age Group	Young adults to mid-career professionals
Education	Higher education / working professionals
Occupation	Knowledge workers, students, professionals producing work artifacts (CVs, reports, drafts) with AI assistance
Digital Literacy	Intermediate–advanced; comfortable with AI tools, but generally hold inaccurate mental models of how memory is stored and retrieved

Attribute	Description
Language	Primarily English
Environment	Web-based chat interface, desktop-first

5. User Persona

Name: Rhea Kapoor (*"The Heavy Personal User"*)

Background: Rhea is a graduate researcher who uses one AI assistant for everything — coursework, a research paper she's writing, and casual personal conversation. She likes that the assistant "remembers" her, but has never been shown what it actually holds about her.

Goals:

- Continuity across sessions without having to re-explain context every time
- Confidence that anything the AI produces on her behalf (like a CV) accurately reflects her current state, not an outdated one

Frustrations:

- Has no idea what specific things the AI has silently decided to remember
- Once had a research paper — still in progress — get recorded by an AI assistant as "completed." That stale memory later propagated into an AI-generated CV, and she was caught out on it in an interview. The memory wasn't wrong when it was written; it just went stale, and nothing ever forced a re-check.
- Has shared sensitive personal context (health, relationship details) in passing and has no idea if or when it might resurface somewhere it doesn't belong

Needs:

- To see and correct what's being remembered *at the moment it's formed*, not buried in a settings page
- A way to mark something "use now, don't keep" with an actual guarantee behind it
- To be asked to confirm memory-derived claims before they're used in something high-stakes, rather than having them asserted as fact

Technical Proficiency: Comfortable using AI tools daily; no understanding of (or interest in) the backend architecture — she just wants the memory to be trustworthy.

5. User Journey

Illustrate how users currently solve the problem and how your solution improves the experience.

Suggested but not definite format:

Stage	Current Experience	Pain Point	Proposed Solution
1. Onboarding & Consent	User starts chatting; memory behavior is undisclosed or buried in terms.	No clear choice about whether memory is even on.	Three-step onboarding (what memory is → disclaimer → explicit opt-in/decline); declining is one click and immediately flips every message to off-the-record.

Stage	Current Experience	Pain Point	Proposed Solution
2. Everyday Conversation	User mentions ongoing work ("writing a paper"), personal notes, casual details.	Extraction happens invisibly; user can't predict what gets kept.	Candidate memories surface as inline review cards after the turn; user accepts, edits, or rejects each one — high-confidence/low-sensitivity items pass quietly to protect an "interruption budget."
3. Time Passes	The paper gets finished, but the AI still "remembers" it as in-progress (or vice versa).	**Staleness** — memory records a moment as if permanent.	Every memory item carries an assertion status (in_progress, completed, etc.), dated rather than treated as timelessly true; stale items are surfaced as "last I knew…" instead of stated as current.
4. High-Stakes Output	User asks the AI to draft a CV using what it remembers.	The stale "in progress" paper is written up as "published" — an actual incident that happened to our own team.	A stake-proportional verification pass checks every memory-derived claim in the draft against its real assertion status in code, and surfaces unconfirmed claims for the user to confirm instead of asserting them.
5. Sensitive Disclosure	User mentions a health detail or shares a card number mid-chat.	No warning before irreversible data leaves the device; sensitive info persists indefinitely with no context boundary.	Client-side PII scan hard-blocks irreversible data before send (redact-or-send-as-is); sensitive-but-legitimate info defaults to session-only storage automatically.
6. Review & Control	User has to dig through a settings menu to see or delete anything.	Memory only auditable after the fact, if at all.	Coequal list/graph memory view; deleting an item previews what else it affects, as text, before confirming; nothing is remembered that can't be traced to the message it came from.

5. Design Process

User assumptions: We assumed AI memory fails less on raw accuracy than on *temporal validity* — a fact can be true when written and wrong later, and nothing in typical memory systems forces a re-check. This assumption came

from an incident within our own team: a research paper recorded as "complete" while still in progress propagated into an AI-generated CV and was caught in an interview.

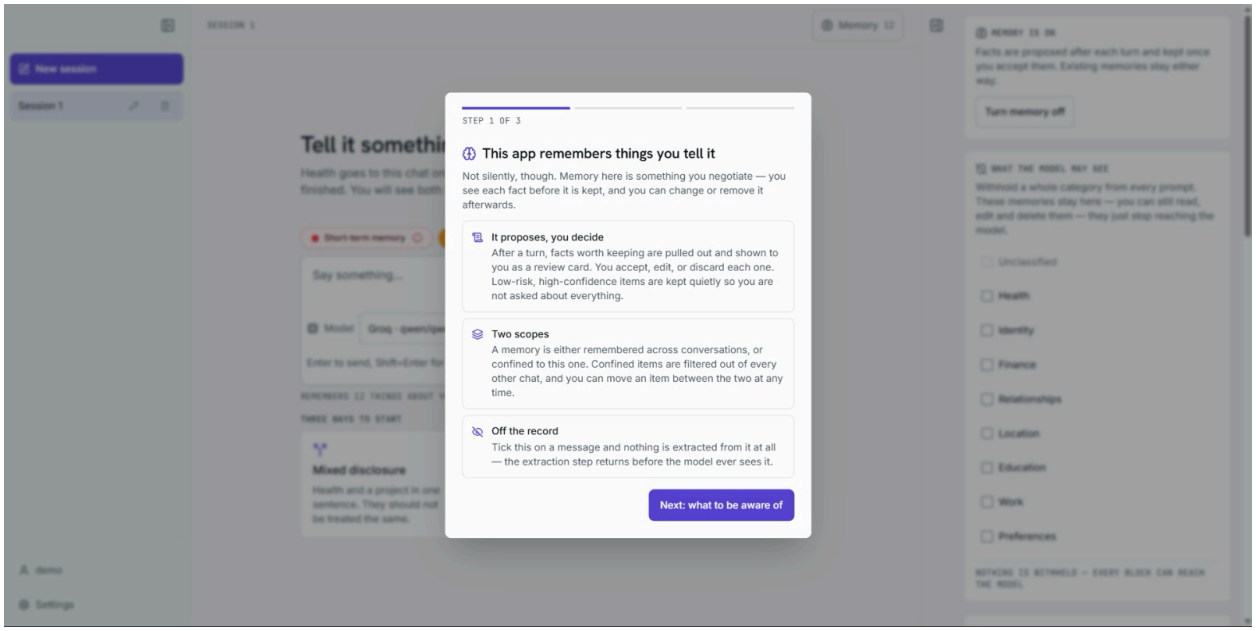
Research and observations: We grounded the problem space in five CHI-track papers on AI memory (transparency needs, expectancy violations, source-memory failure, agentic memory design) and ran a 34-response survey confirming the pattern from both directions — 58.8% of respondents had to repeat information within a single conversation, while 47.1% had a moment the AI *should* have remembered and didn't.

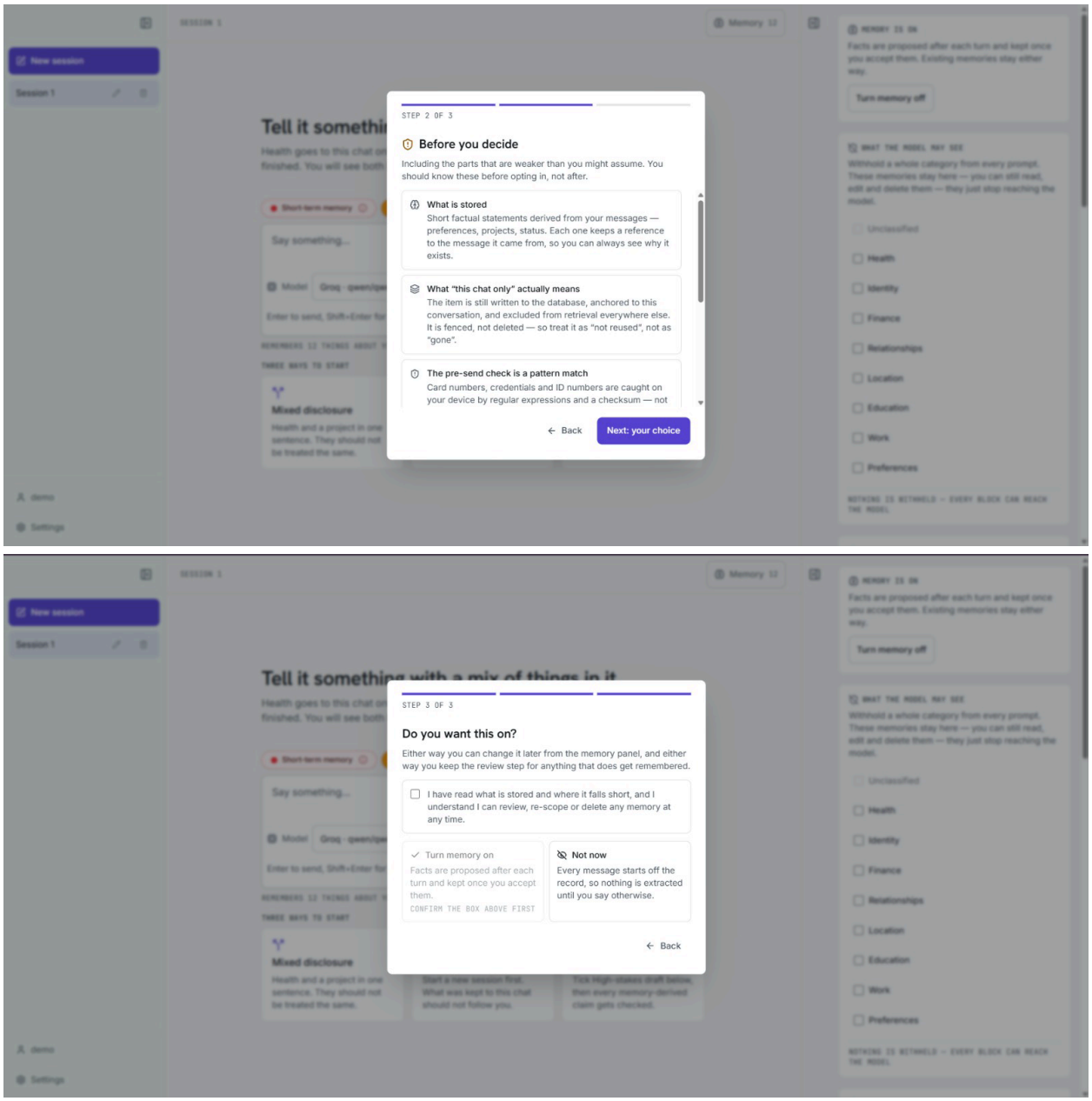
Brainstorming: Early concepts ranged from a pure "memory dashboard" to fully automatic silent memory. Both were rejected as anti-goals: a dashboard treats memory as something to audit rather than negotiate; silent memory offers no moment of user control.

Concept selection: We converged on negotiation-at-formation-time: memory candidates surface inline, immediately, and are decided on before they persist — never stored first and reviewed later.

Iterations: The memory panel began as a fixed 50/50 split with the transcript and was collapsed into a toggleable column (D42→D44) after we found it competed with the conversation at moments that mattered most, like a live demo. The graph view was moved to a full-page tab reusing the same underlying component as the list (D46), so no capability is reachable *only* through the graph. The color palette was re-audited for contrast after a swap, catching three tokens that had silently become invisible or semantically inverted (D45).

7. Interface Design





Figs 1, 2, 3 : User On-Boarding for Memory Feature

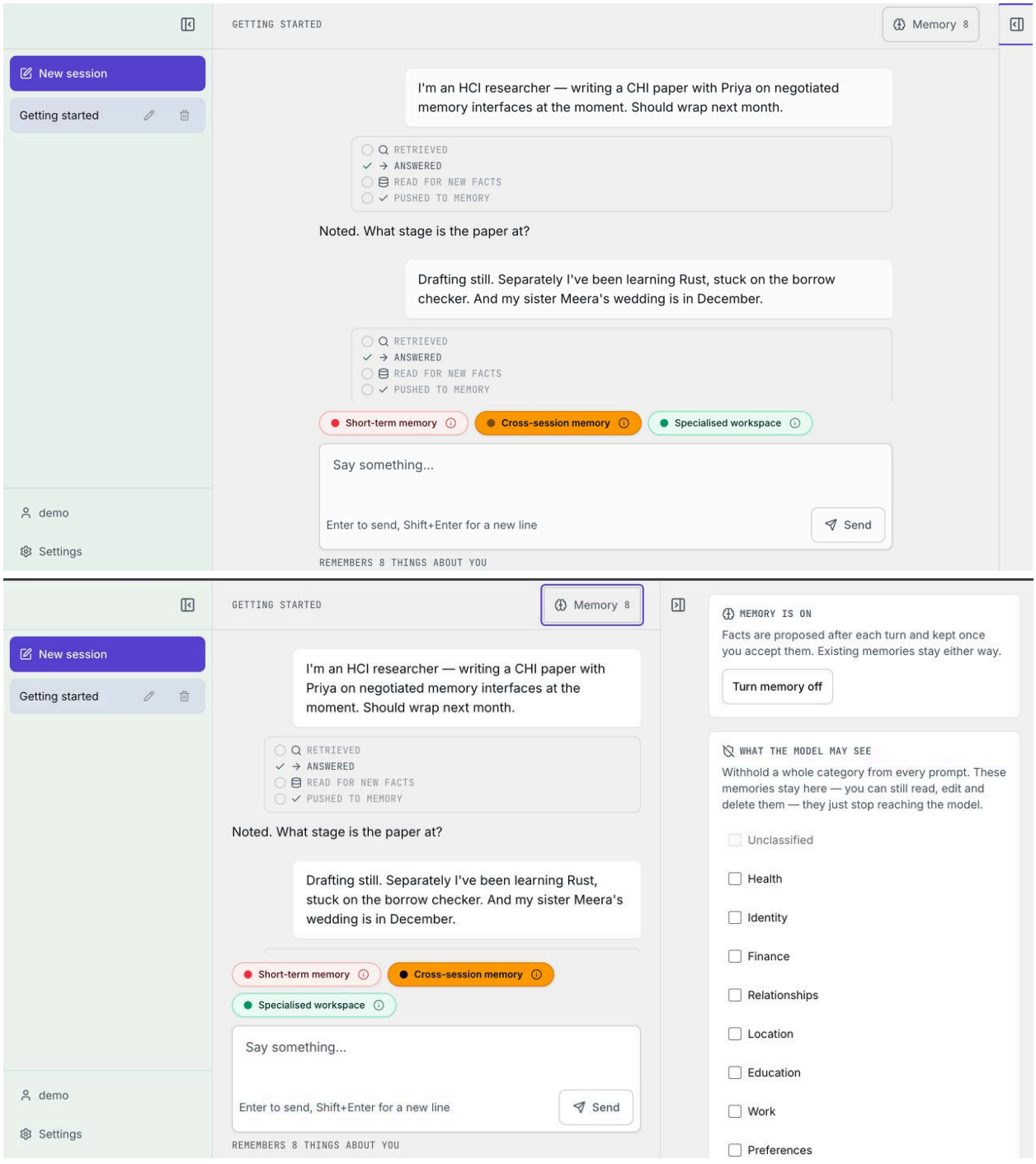


Fig 4,5 : Transparency in Data Usage and Retrieval

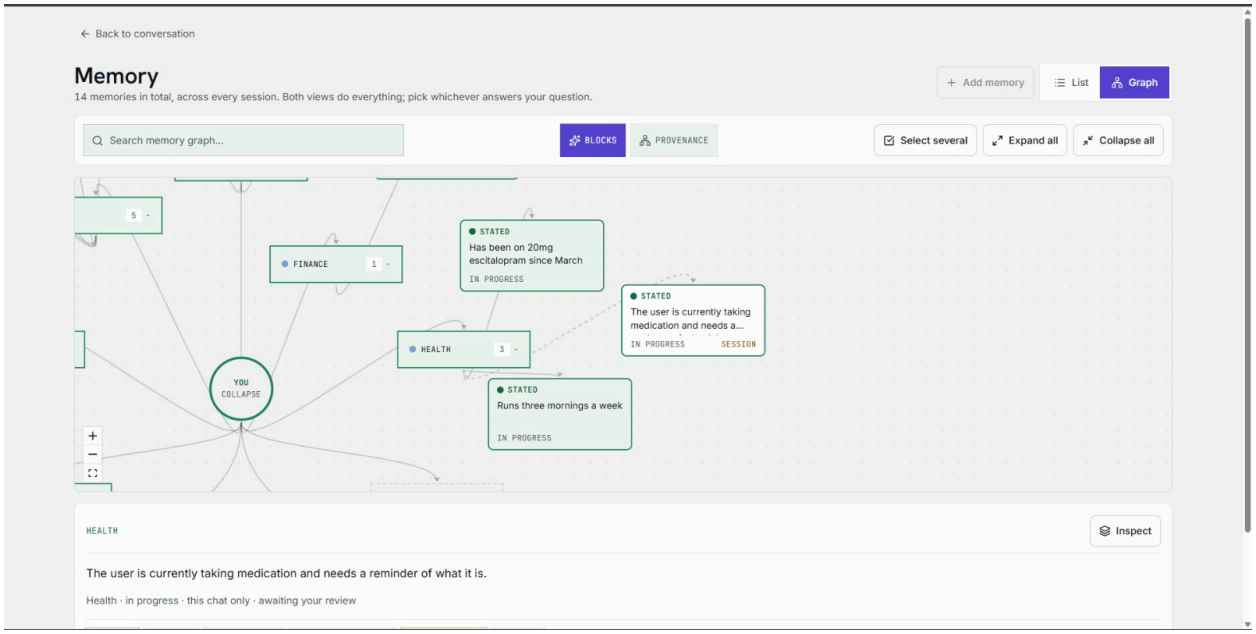


Fig 6 : Memory Graph with Auto-Categorization

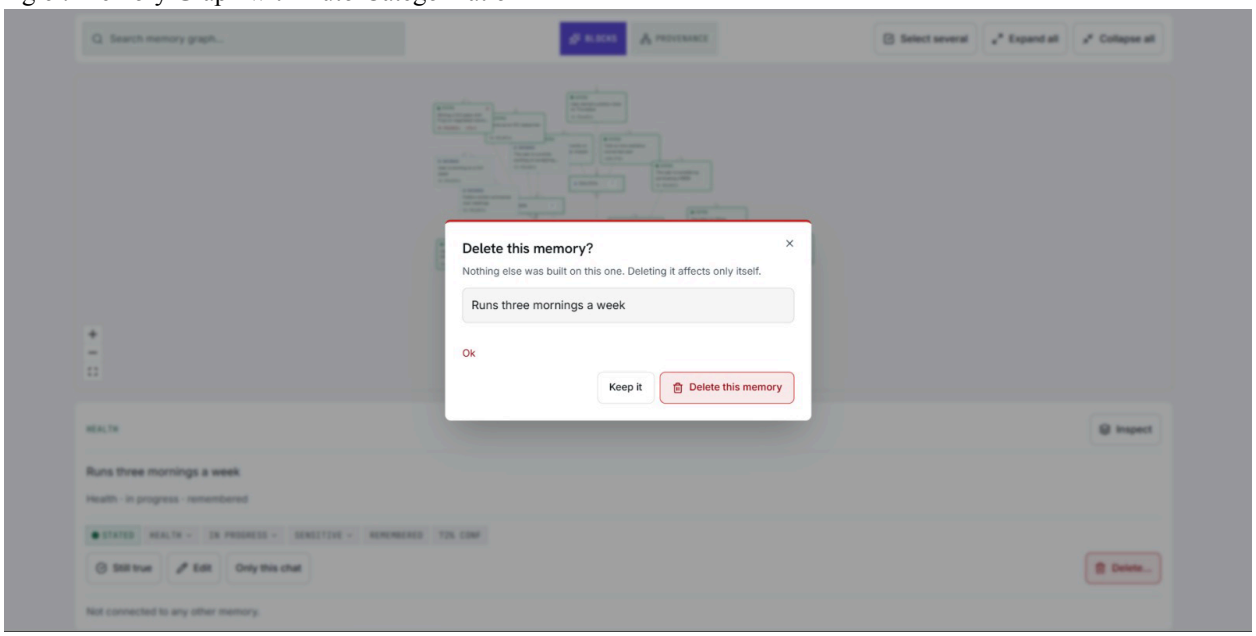
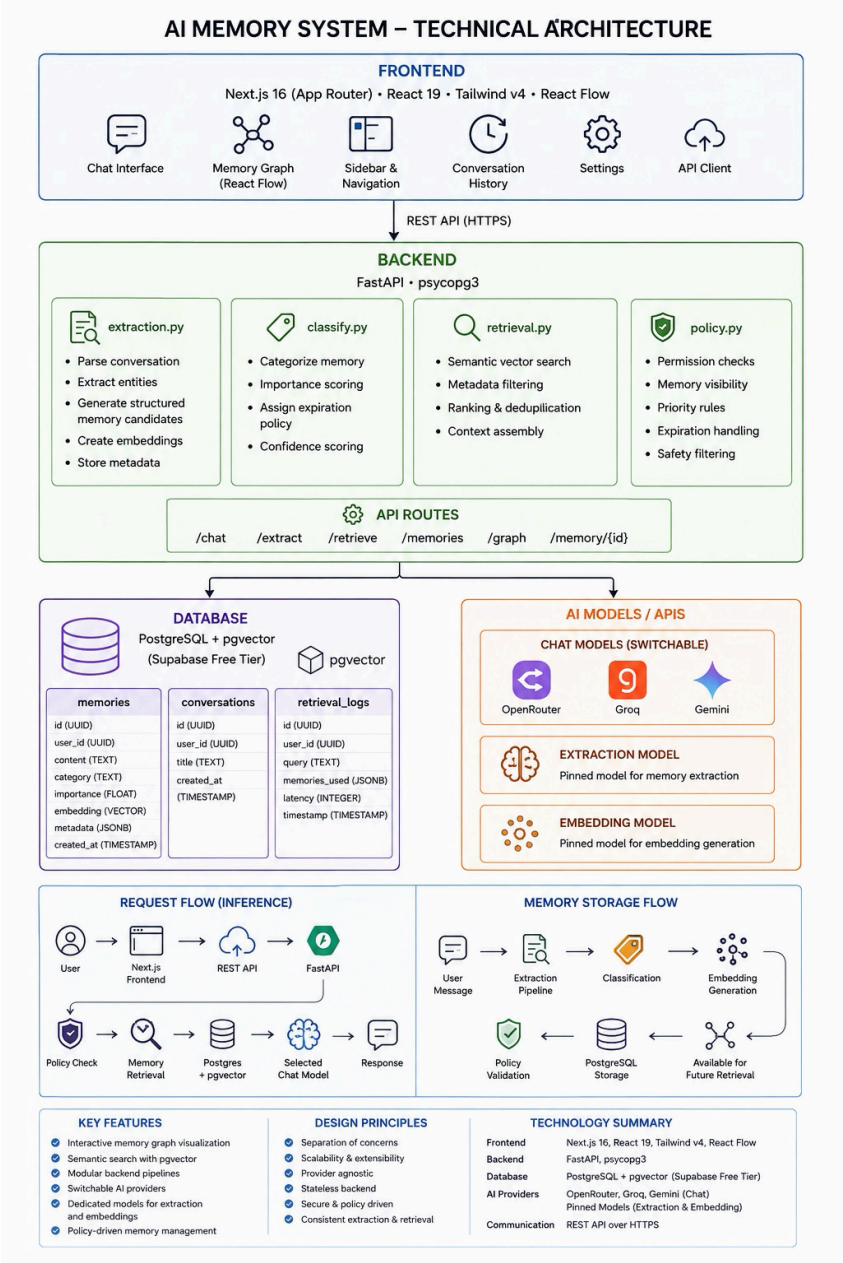


Fig 7 : Control over CRUD Operations by user

11. Technical Architecture

- **Frontend:** Next.js 16, React 19, Tailwind v4 (CSS-first @theme)
- **Backend:** FastAPI + psycopg3
- **Database:** Postgres + pgvector
- **AI Models/APIs:** chat is switchable per-turn (OpenRouter, Groq, Gemini); extraction, classification, and embeddings are pinned to one model regardless of chat provider, so switching chat mid-demo can't silently fragment the vector space or make status classification non-reproducible
- **External APIs:** OpenRouter/Groq/Gemini for inference; response always reports which provider *actually* answered, with fallback if a slug is retired.



14. Results and Discussion

Who benefits? Heavy personal users gain continuity without giving up visibility or control over what's remembered. Cautious professionals benefit most from the stake-proportional verification pass — it directly catches the failure mode that motivated this project: a stale memory silently becoming a false claim in a high-stakes artifact like a CV.

How does it improve existing workflows? Most memory systems expose control only *after* storage, via a settings page. Ours moves the decision point to the moment memory forms — before anything persists. Several of the strongest claims (the session-only guarantee, consent gating extraction) are verified structurally, not just asserted in the UI — e.g., a direct database query proving zero rows persist from an off-the-record turn.

Can it scale? Yes, by design. Extraction and classification are decoupled from the chat provider, so new chat models plug in without touching the memory pipeline. The assertion-status schema is a generalizable pattern beyond this use case — any system needing a decay-aware notion of "this was true when written" could reuse it. Real multi-user scaling is an explicitly named gap, not a hidden one.

15. Future Work

- Bounded relational transparency — engineer relational cues into the assistant's presentation, while keeping "scrutiny" structural: a short, periodic profile summary surfaced to the user (not buried in settings), with a standing invitation to confirm, correct, or delete the assumptions it's built up
- Cross-session persona consistency — extend the relational-cue work to check that the "Me Memory" the assistant presents stays coherent across sessions, rather than drifting as new facts get added.

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Appendix (Optional)

1. Survey Data:



2. Affinity Mapping:

